Applied Mathematics for Chemists
(화학수학)

Lecture 18

Group Theory

Trace of a Square Matrix (대각합)

$$\mathrm{tr}\,\mathbf{A} = \sum_{i=1}^{n} a_{ii}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \mathrm{tr}\,\mathbf{A} = a_{11} + a_{22} + a_{33}$$

Even when $\mathbf{AB} \neq \mathbf{BA}$,

$$\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$$

Determinant of a Square Matrix (행렬식)

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

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Applications of the Determinant

- Slater determinant
- Vector product of two vectors
- Jacobian calculation
- Determinant of a singular matrix
- Matrix eigenvalue problem

Matrix Eigenvalue Problem

For a given square matrix $\mathbf{B}$, if we have

$$\mathbf{B}\vec{x} = b\vec{x}$$

- $\vec{x}$ (column vector): eigenvector
- b (scalar): eigenvalue

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

$$\det(\mathbf{B} - b\mathbf{E}) = 0$$

Today's Class

- Groups
  - Point group
  - Representation of a point group
  - Characters
  - Irreducible representations
  - Character tables

Group (군)

Collection of members

with specific mathematical properties

Useful for studying systems with symmetry

Applications

Molecular orbitals of polyatomic molecules

Spectroscopic selection rules

Requirements for a Group

1. If A and B are members of the group, then the product AB is also a member of the group
2. The group must contain the identity E such that

$$AE = EA = A \text{ (for every member } A\text{)}$$

3. The inverse of every member A^{-1} is a member

$$A^{-1}A = AA^{-1} = E$$

4. The associative law holds: $A(BC) = (AB)C$

Example

group of $\{1,\ i,\ -1,\ -i\}$

- Multiplication table

	$\mathbf{1}$	$\boldsymbol{i}$	$\mathbf{-1}$	$\boldsymbol{-i}$
$\mathbf{1}$	1	i	-1	$-i$
$\boldsymbol{i}$	i	-1	$-i$	1
$\mathbf{-1}$	-1	$-i$	1	i
$\boldsymbol{-i}$	$-i$	1	i	-1

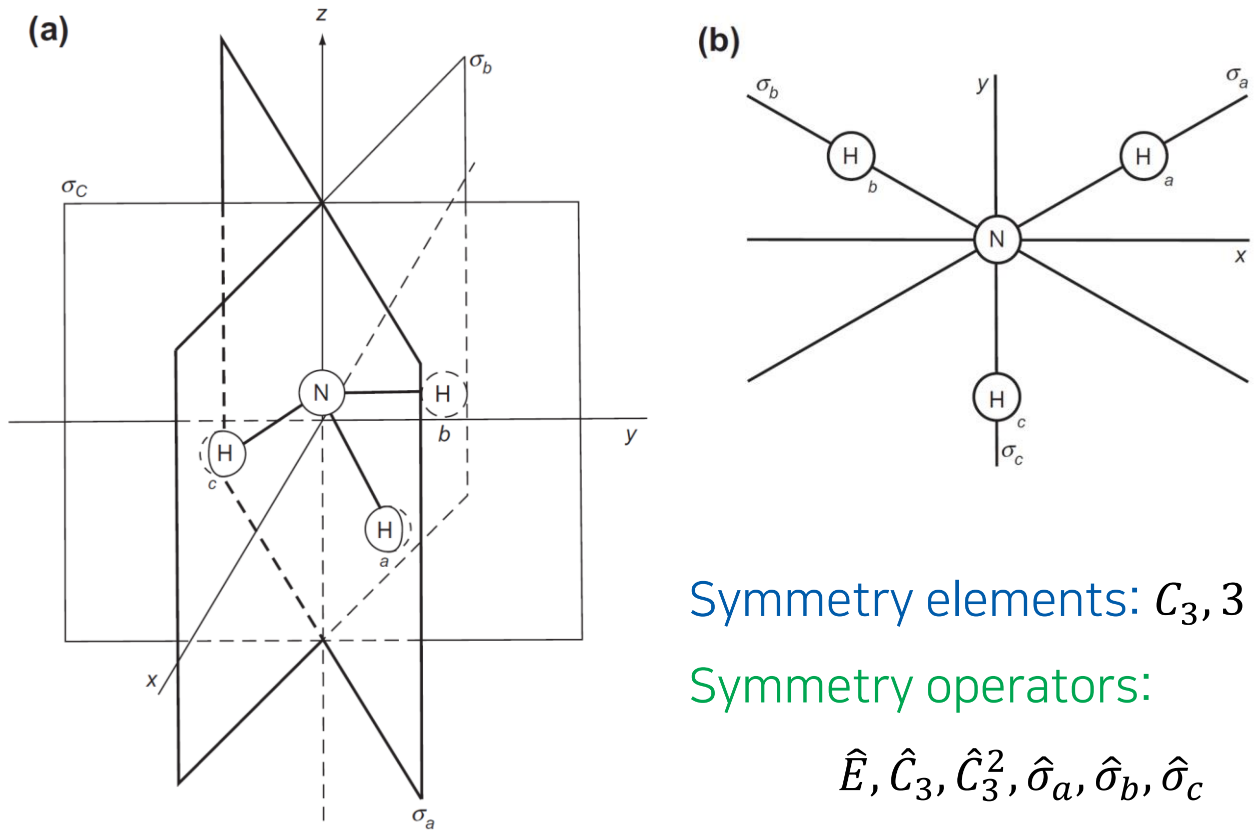
Identity = 1

Point Group (점군)

Formed by a set of point symmetry operators

belonging to a symmetrical object

Example: Ammonia (NH_3)



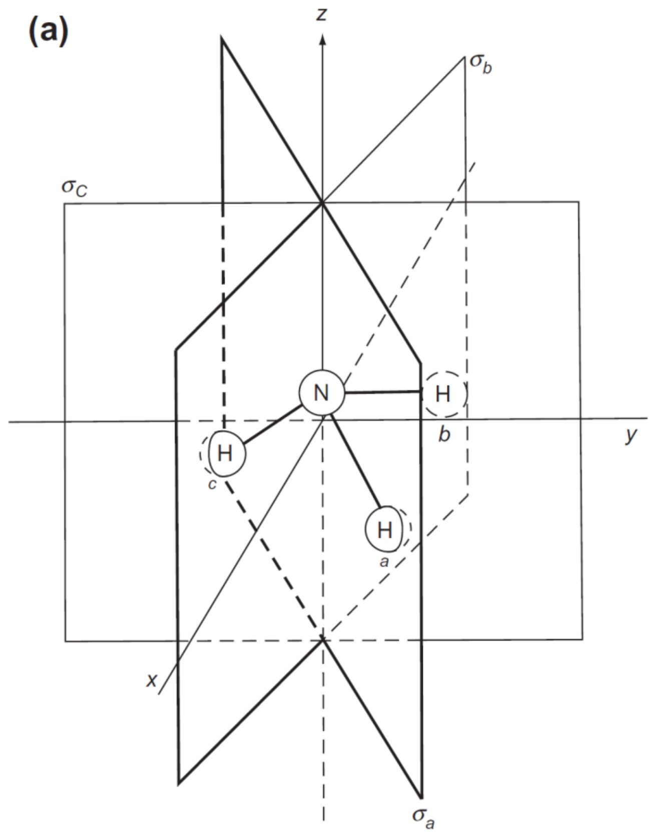
Symmetry elements: $C_3, 3 \times \sigma_\mathrm{v}$

Symmetry operators:

$$\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c$$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.4)

Example: Ammonia (NH_3)



Symmetry elements: $C_3, 3 \times \sigma_\mathrm{v}$

Symmetry operators:

$$\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c$$

Multiplication table:

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.4)

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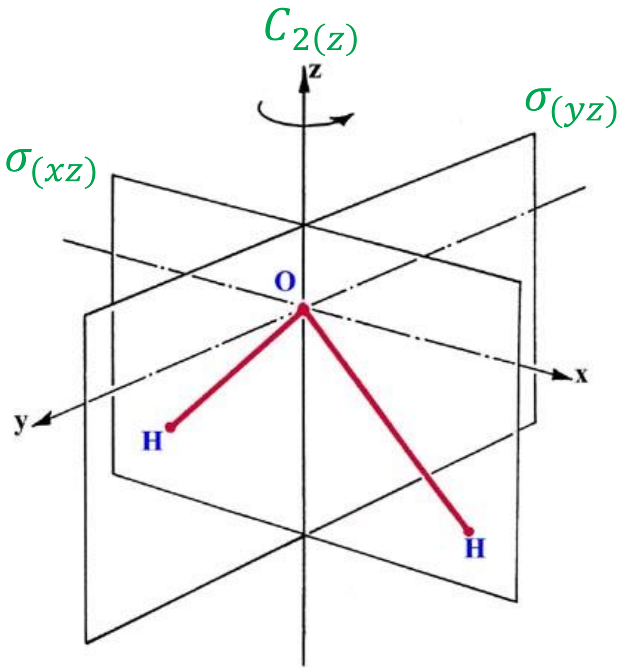
Finding a Multiplication Table

if you have a reflection plane that includes all atoms…



(Image: https://chem.libretexts.org/@go/page/64756)

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Schoenflies Symbol

Each point group is assigned a symbol

Symbol	Symmetry elements
C_1	E
C_s	E, σ
C_i	E, i
C_n	E, C_n
S_n	E, S_n
C_{nv}	$E, C_n, n \times \sigma_\mathrm{v}$
C_{nh}	$E, C_n, \sigma_\mathrm{h}$
D_n	$E, C_n, n \times C_2$
D_{nh}	$E, C_n, n \times C_2, \sigma_\mathrm{h}$
D_{nd}	$E, C_n, n \times C_2, n \times \sigma_\mathrm{v}$
T_d	$E, 6 \times \sigma_\mathrm{d}, 4 \times C_3, 3 \times S_4$
O_h	$E, 6 \times \sigma_\mathrm{d}, 3 \times \sigma_\mathrm{h}, 6 \times C_2, 4 \times C_3, 3 \times C_4, 3 \times S_4, 4 \times S_6$

Representation of a Point Group

Symmetry operator ↔ Matrix

$$\hat{A}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\mathbf{A}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

Identity Symmetry Operator

$$\hat{E} \;\leftrightarrow\; \mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

How to Find the Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}y_1 + a_{13}z_1 = x_2 \\ a_{21}x_1 + a_{22}y_1 + a_{23}z_1 = y_2 \\ a_{31}x_1 + a_{32}y_1 + a_{33}z_1 = z_2 \end{cases}$$

From the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2), determine a_{ij}

Example 13.20

Find the matrix that represents $\hat{C}_{n(z)}$.

The operation transforms (x_1, y_1, z_1) to (x_2, y_2, z_2) as follows:

$$x_2 = \left(\cos\frac{2\pi}{n}\right)x_1 - \left(\sin\frac{2\pi}{n}\right)y_1$$

$$y_2 = \left(\sin\frac{2\pi}{n}\right)x_1 + \left(\cos\frac{2\pi}{n}\right)y_1$$

$$z_2 = z_1$$

Hence, the representation matrix is

$$\hat{C}_{n(z)} \leftrightarrow \begin{bmatrix} \cos(2\pi/n) & -\sin(2\pi/n) & 0 \\ \sin(2\pi/n) & \cos(2\pi/n) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Representation of the C_{3v} Group

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}, \qquad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B}, \qquad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{D}, \qquad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{F}$$

Example 13.21

By matrix multiplication, show that $\mathbf{AB} = \mathbf{E}$.

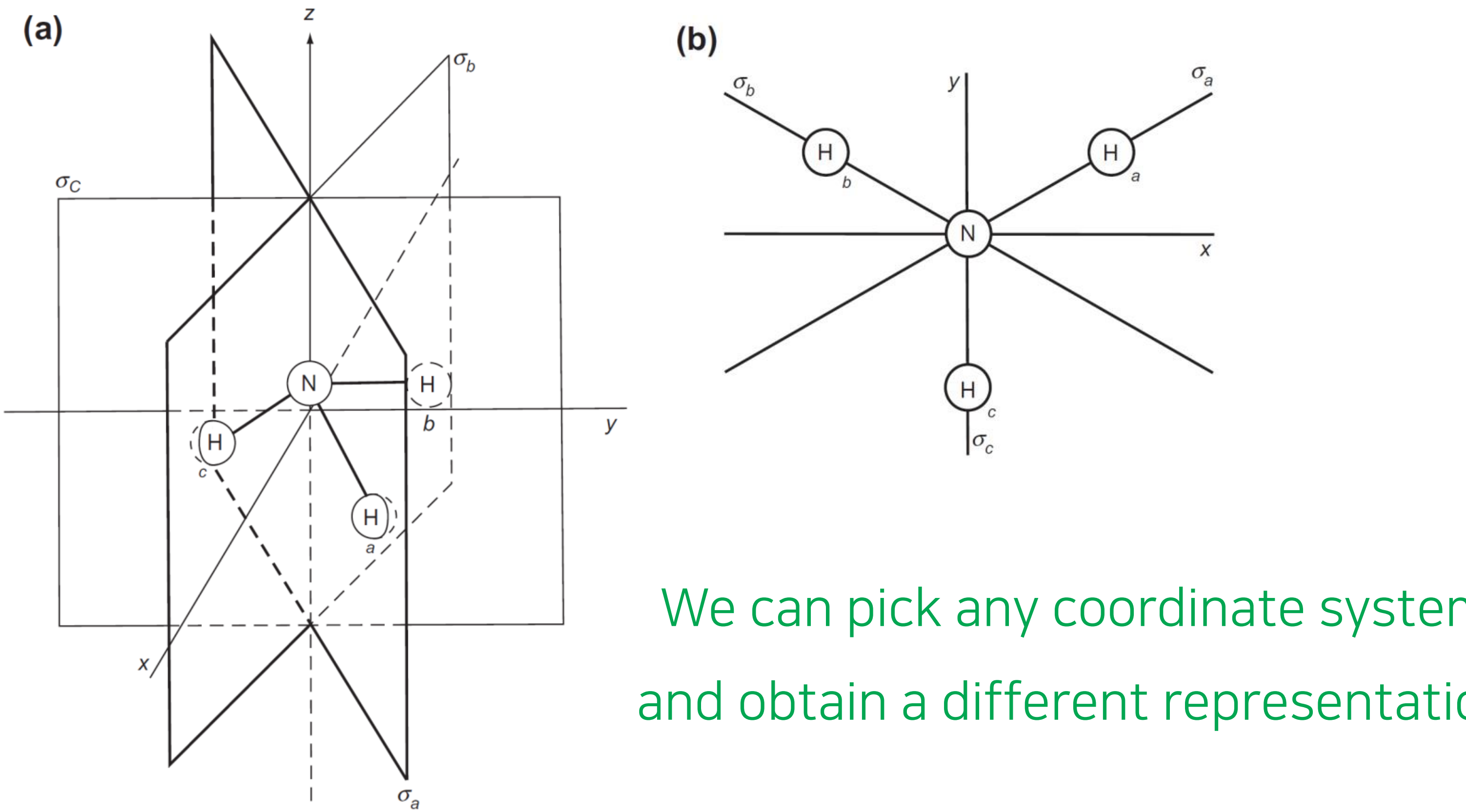
Since

$$\mathbf{A} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

matrix multiplication gives

$$\mathbf{AB} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}$$

Arbitrary Nature of Representation



We can pick any coordinate system and obtain a different representation

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.4)

Similarity Transformation (닮음 변환)

- Two different representations $\mathbf{P}$ and $\mathbf{Q}$ in different coordinates can be linked by

$$\mathbf{P} = \mathbf{X}^{-1}\mathbf{Q}\mathbf{X}$$

when $\mathbf{X}$ is some square matrix

- Since $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$,

$$\mathrm{tr}(\mathbf{P}) = \mathrm{tr}(\mathbf{X}^{-1}\mathbf{Q}\mathbf{X}) = \mathrm{tr}(\mathbf{X}\mathbf{X}^{-1}\mathbf{Q}) = \mathrm{tr}(\mathbf{Q})$$

Character (지표)

- Character of a symmetry operator = trace of the corresponding matrix

$$\hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}$$

Character of $\hat{C}_3 = \mathrm{tr}\,\mathbf{A} = (-1/2) + (-1/2) + 1 = 0$

Our Representation of C_{3v} Group

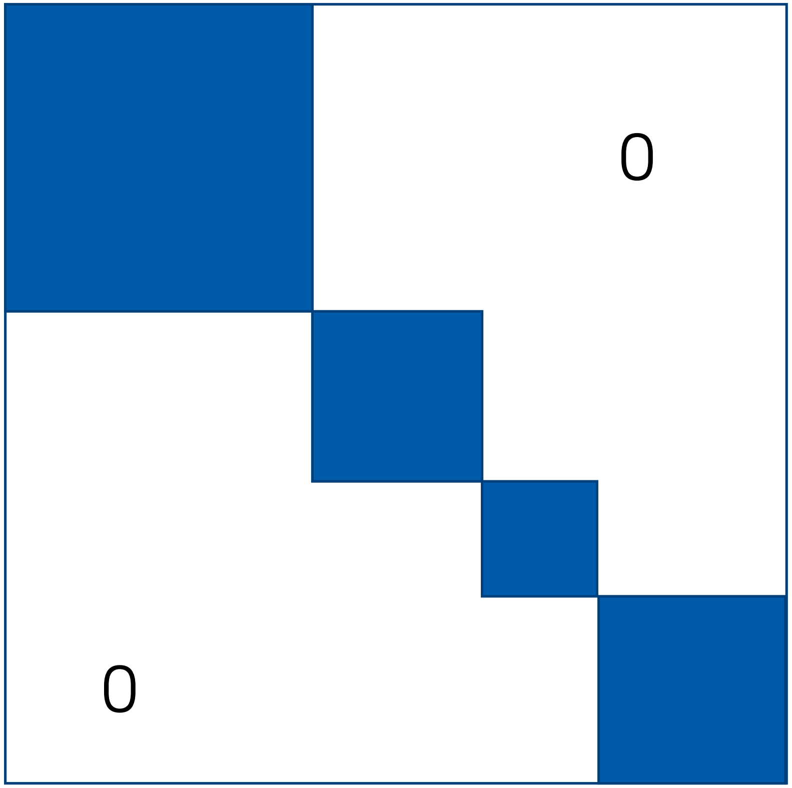
$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}, \qquad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B}, \qquad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{D}, \qquad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{F}$$

$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
3	0	0	1	1	1

Block-Diagonal Matrix (블록대각행렬)



Reducible Representation (가약 표현)

All matrices are block-diagonal, or can be put into block-diagonal form by a similarity transformation

$$\hat{E} \leftrightarrow \left[\begin{array}{cc:c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hdashline 0 & 0 & 1 \end{array}\right] = \mathbf{E}, \qquad \hat{C}_3 \leftrightarrow \left[\begin{array}{cc:c} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ \hdashline 0 & 0 & 1 \end{array}\right] = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \left[\begin{array}{cc:c} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ \hdashline 0 & 0 & 1 \end{array}\right] = \mathbf{B}, \qquad \hat{\sigma}_a \leftrightarrow \left[\begin{array}{cc:c} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ \hdashline 0 & 0 & 1 \end{array}\right] = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \left[\begin{array}{cc:c} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ \hdashline 0 & 0 & 1 \end{array}\right] = \mathbf{D}, \qquad \hat{\sigma}_c \leftrightarrow \left[\begin{array}{cc:c} -1 & 0 & 0 \\ 0 & 1 & 0 \\ \hdashline 0 & 0 & 1 \end{array}\right] = \mathbf{F}$$

Block Submatrices

- Block submatrices obey the same multiplication table of the group of symmetry operators

- Corresponding block submatrices can form a new representation

Example: 2 by 2 Blocks

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \qquad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \qquad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

Example: 1 by 1 Blocks

$$\hat{E} \leftrightarrow 1, \qquad \hat{C}_3 \leftrightarrow 1, \qquad \hat{C}_3^2 \leftrightarrow 1,$$

$$\hat{\sigma}_a \leftrightarrow 1, \qquad \hat{\sigma}_b \leftrightarrow 1, \qquad \hat{\sigma}_c \leftrightarrow 1$$

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

Irreducible Representation (기약 표현)

Cannot be made block-diagonal

by any similarity transformation

- Reduction of a representation (표현의 약분): to make the matrices in the representation block-diagonal by some similarity transformation
- Often abbreviated as irrep

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Examples of C_{3v}

- 2 by 2 blocks: one irrep

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \qquad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \qquad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- 1 by 1 blocks: another irrep

$$\hat{E} \leftrightarrow 1,\ \hat{C}_3 \leftrightarrow 1,\ \hat{C}_3^2 \leftrightarrow 1,\ \hat{\sigma}_a \leftrightarrow 1,\ \hat{\sigma}_b \leftrightarrow 1,\ \hat{\sigma}_c \leftrightarrow 1$$

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Characters of Irreps

Class:	E	$2C_3$		$3\sigma_v$		
Irrep	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
2 by 2	2	-1	-1	0	0	0
1 by 1	1	1	1	1	1	1

Class is constituted of symmetry operations

that can be converted by similarity transformations

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Character Table (지표표)

- Using group theory theorems, we can find all possible irreps and collect their characters → character table
- *e.g.* C_{3v} group

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

conventional notation for irreps

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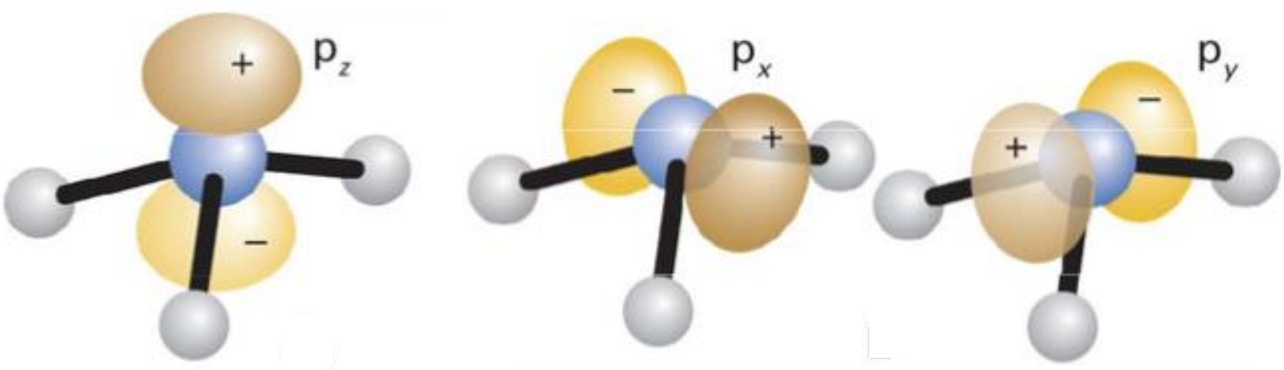
Application of Character Table

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0



(Image: https://commons.wikimedia.org/wiki/File:Orbitals_of_NH3.png)